RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 1

Title: Creating an AWS Account, Exploring AWS Global Infrastructure, and Configuring AWS CLI

1. Learning Objectives:

After completing this experiment, the student will be able to:

- Create and activate an Amazon Web Services (AWS) account
- Understand AWS global infrastructure concepts such as Regions, Availability Zones, and Edge Locations
- Navigate the AWS Management Console
- Install and configure AWS Command Line Interface (CLI)
- Verify AWS CLI configuration using basic commands

2. Theory:

Amazon Web Services (AWS):

Amazon Web Services (AWS) is a secure cloud services platform that provides computing power, storage, databases, networking, and many other services on a pay-as-you-go basis. AWS helps organizations reduce infrastructure cost and scale applications globally.

AWS Global Infrastructure:

AWS operates a globally distributed infrastructure designed for high availability, fault tolerance, and low latency.

- Regions:
  A Region is a geographical area that contains multiple Availability Zones. Each region is isolated from others to ensure data security and compliance.
  Example: ap-south-1 (Mumbai), us-east-1 (N. Virginia)
- Availability Zones (AZs):
  Availability Zones are physically separate data centers within a region, connected through low-latency links. AZs help design highly available and fault-tolerant applications.
- Edge Locations:
  Edge Locations are used by AWS services such as Amazon CloudFront to deliver content with low latency to end users by caching data closer to them.

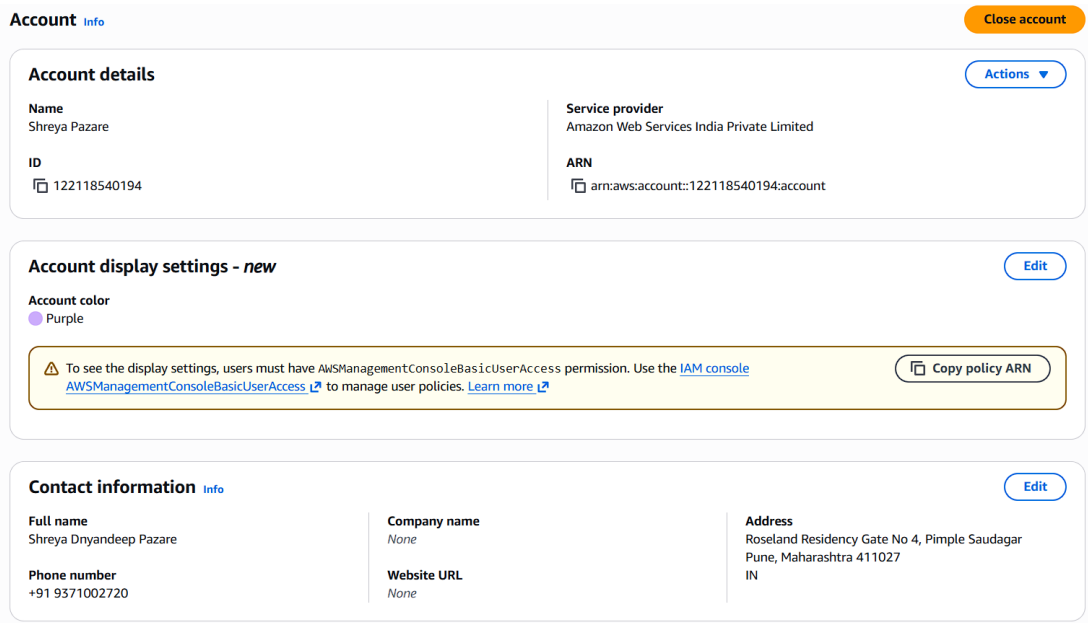
AWS Command Line Interface (CLI):

AWS CLI is a unified command-line tool that allows users to interact with AWS services using commands. It enables automation, scripting, and faster resource management compared to the web console.

3. Procedure:

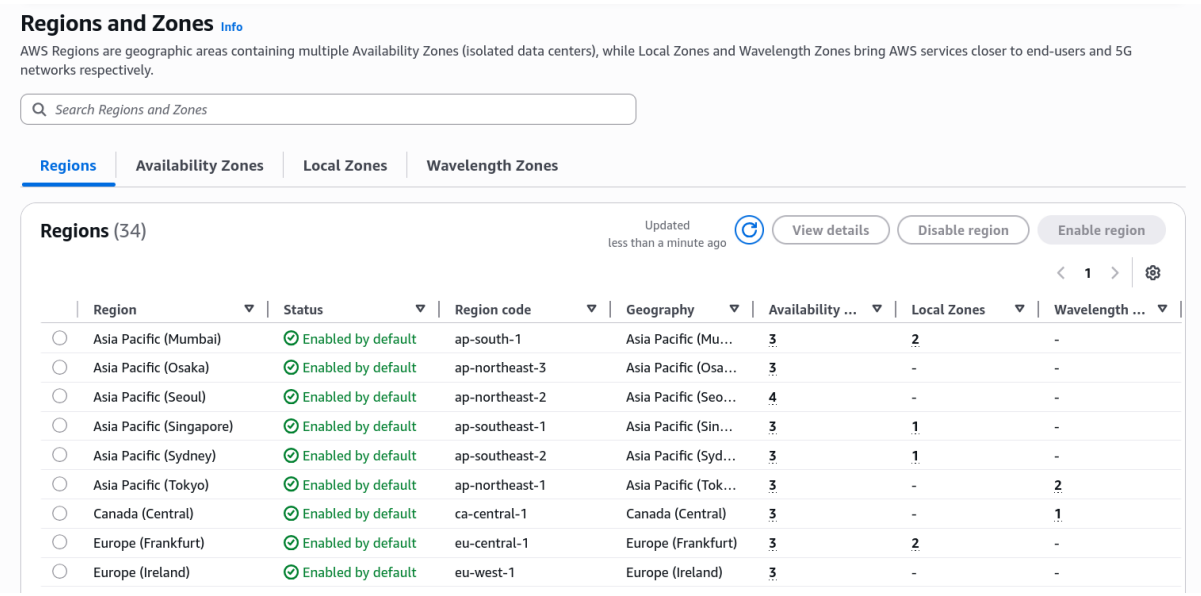
Step 1: Create an AWS Account

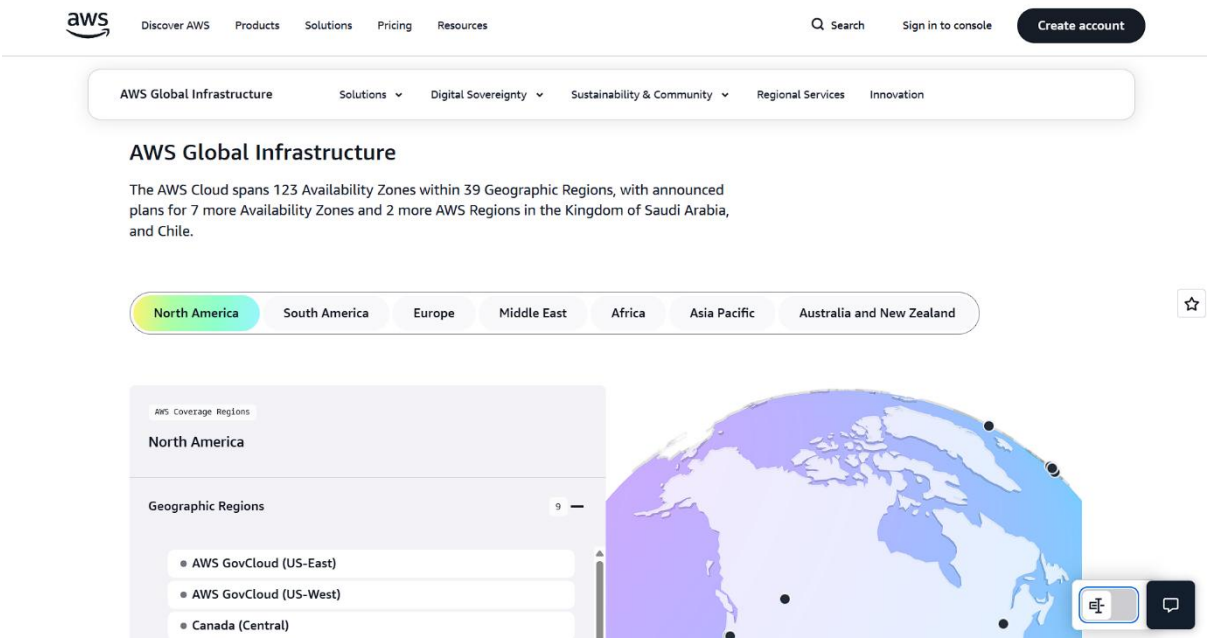
1. Open a web browser and go to https://aws.amazon.com/
2. Click on Create an AWS Account
3. Enter email address, password, and AWS account name
4. Provide contact information and billing details
5. Complete identity verification
6. Select Basic Support Plan
7. Log in to the AWS Management Console



Step 2: Explore AWS Global Infrastructure

1. Log in to AWS Management Console
2. From the top-right corner, click on the Region selector
3. Observe different regions available globally
4. Navigate to AWS Global Infrastructure page
5. Identify:
    - Number of Regions
    - Availability Zones per Region
    - Edge Locations worldwide





Step 3: Install AWS CLI

1. Download AWS CLI from: https://aws.amazon.com/cli/
2. Install AWS CLI based on your operating system (Windows/Linux/Mac)
3. Open Command Prompt or Terminal
4. Verify installation using:
5. aws --version

```
~
→ aws --version
aws-cli/2.34.19 Python/3.14.3 Linux/6.6.87.2-microsoft-standard-WSL2 exe/x86_64.ubuntu.22
```

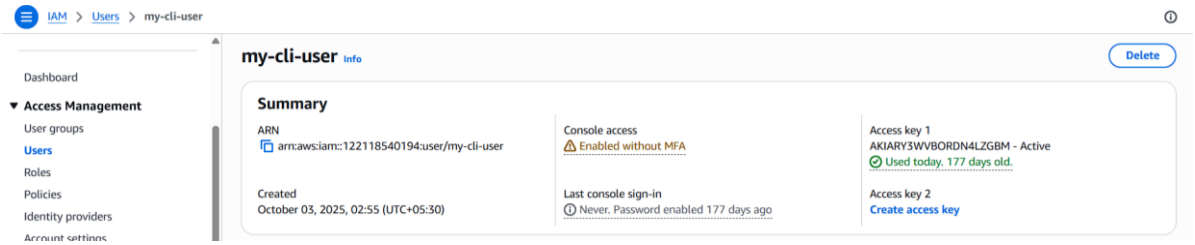
Step 4: Configure AWS CLI

1. Generate Access Keys:
   - Go to IAM → Users → Security Credentials
   - Create Access Key and Secret Access Key
2. Open terminal or command prompt
3. Run the following command:

   aws configure

4. Enter:
   - AWS Access Key ID

- AWS Secret Access Key
- Default region name (e.g., ap-south-1)
- Default output format (json)



Step 5: Verify AWS CLI Configuration

1. Execute the command:
2. aws sts get-caller-identity
3. Confirm that account details are displayed successfully

```
→ aws sts get-caller-identity
{
    "UserId": "AIDARY3WVBOREQATD27EV",
    "Account": "122118540194",
    "Arn": "arn:aws:iam::122118540194:user/my-cli-user"
}
```

4. Questions:

1. What is AWS and why is it used?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform. Instead of buying, owning, and maintaining physical data centers and servers, you "rent" technology services—such as computing power, storage, and databases—on an as-needed basis.

Why is it used?

- Cost Efficiency: You only pay for what you use (the "Pay-as-you-go" model). No more paying for idle servers.
- Scalability & Elasticity: You can start small and instantly scale up to handle millions of users, then scale back down to save money.
- Global Reach: You can deploy your application to users around the world in minutes.
- Security: AWS handles the heavy lifting of physical security and infrastructure compliance.

2. What is an AWS Region?

An AWS Region is a physical location in the world where Amazon clusters multiple data centers. Each Region is a separate geographic area (for example, us-east-1 in North Virginia or ap-south-1 in Mumbai).

Regions are completely independent of one another. This ensures that even if a natural disaster hits one part of the world, your data in another Region remains safe.

3. Region vs. Availability Zone (AZ)

Think of a Region as a city and an Availability Zone as a specific building or campus within that city.

Feature	AWS Region	Availability Zone (AZ)
Definition	A geographic area (e.g., US East).	One or more discrete data centers within a Region.
Isolation	Completely independent and isolated.	Isolated from other AZs but connected via high-speed fiber.
Purpose	Data sovereignty and low-latency for local users.	Fault tolerance and High Availability (HA).
Example	us-east-1	us-east-1a, us-east-1b

4. What command is used to verify AWS CLI configuration?

If you want to make sure your credentials and default region are set up correctly on your local machine, use the following command:

Bash:

aws configure list

What this shows:

- Access Key: The ID you're using.
- Secret Key: (Masked for security).
- Region: The default location your commands are sent to.

If you want to verify that your credentials actually *work* and have permission to talk to AWS, run:

aws sts get-caller-identity